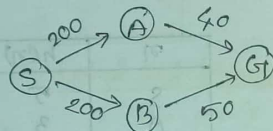


A* Admissibility

$h(n) \leq h^*(n) \rightarrow$ underestimation

$h(n) > h^*(n) \rightarrow$ overestimation

↓
Estimated cost Actual cost



n	$h(n)$
S	5
A	
B	
G	0

For overestimation

$$h(A) = 80$$

$$h(B) = 70$$

Initiation $f(S) = 0 + 5 = 5$

Iteration 1:

$$\begin{array}{ll}
 S \rightarrow A & S \rightarrow B \\
 200 + 80 = 280 & 200 + 70 = 270
 \end{array}$$

Iteration 2:

$$\begin{array}{ll}
 S \rightarrow B \rightarrow G & S \rightarrow A \\
 200 + 50 + 0 = 250 & 200 + 80 = 280
 \end{array}$$

$\therefore$ path = $S \rightarrow B \rightarrow G$

For underestimation)

$$h(A) = 30$$

$$h(B) = 20$$

Initiation $f(S) = 0 + 5 = 5$

Iteration 1:

$$S \rightarrow A$$

$$S \rightarrow B$$

$$200 + 30 = 230 \quad 200 + 20 = 220$$

Iteration 2:

$$S \rightarrow B \rightarrow G$$

$$S \rightarrow A$$

$$200 + 50 + 0 = 250 \quad 200 + 30 = 230$$

Iteration 3:

$$S \rightarrow A \rightarrow G$$

$$S \rightarrow B \rightarrow G$$

$$200 + 40 + 0 = 240 \quad 200 + 50 + 0 = 250$$

A* search provide an optimal solution to underestimation case. So we can say that A* search algorithm is admissible.

Propositional logic / Boolean

- ✓ Propositional logic is the declarative statement either true or false. (ছব্বী সত্য বা মিথ্যে) not both.
- ✓ Tautology: If the declarative statement is always true
- ✓ Contradiction: If the declarative statement is always false

Proposition logic type

1. atomic $\rightarrow$ simple statement is atomic
2. compound $\rightarrow$ two or more atomic proposition

logical connectives (operators)

1. Negation ($\neg$)
2. Conjunction ($\wedge$)
3. Disjunction ($\vee$)
4. Implication ($\rightarrow$) if then
5. Biconditional ($\leftrightarrow$) iff

P	Q	$P \wedge Q$	$P \vee Q$	$P \rightarrow Q$	$P \leftrightarrow Q$
T	T	T	T	T	T
T	F	F	T	F	F
F	T	F	T	T	F
F	F	F	F	T	T

logical equivalence $A \rightarrow B = \neg A \vee B$

Properties of operators

1. Commutativity $\rightarrow p \vee q = q \vee p$
2. Associativity $\rightarrow (p \vee q) \vee r = p \vee (q \vee r)$
3. Identity element $\rightarrow p \wedge \text{true} = p$
4. Distributive $\rightarrow p \vee (q \wedge r) = (p \vee q) \wedge (p \vee r)$
5. De Morgan's law $= \neg(A \wedge B) = \neg A \vee \neg B$
6. Double-negation elimination $= \neg(\neg A) = A$

Limitation:

(all) girls are intelligent
(some) students are missing
Don't allow propositional logic
to solve this we use Predicate logic

Wumpsa world:

4	stench		Breeze	PIT
3		stench Breeze	PIT	breeze
2	stench		Breeze	
1		Breeze	PIT	breeze
	1	2	3	4

knowledge-base-agent.

knowledge base agents are those agent who have the capacity / capability of maintaining a internal state of knowledge. reason over the knowledge, update their knowledge after observation. and take action.

- * knowledge

- * Decision making

- * reasoning

- * Act

✓ operation performed by knowledge base Agent:

1. Tell : The operation Tell the KB what it perceives from the env.
2. Ask : what action it should perform
3. Perform : It perform the selected action

Function KB-AGENT($percept$);

Persistent : KB, a knowledge base
 t , a counter, initially 0, indicating time

```
TELL(MAKE-PERCEPT-SENTENCE( $percept$ ,  $t$ ))  
ACTION = ASK(KB, MAKE-ACTION-QUERY( $t$ ))  
TELL(KB, MAKE-ACTION-SENTENCE( $action$ ,  $t$ ))  
 $t = t + 1$   
return action
```


Masud Sir

CH-18

Learning from examples

- ✓ * Define supervised and unsupervised learning.
writedown the algorithm of forward-propagation (4)
- ✓ * Define machine learning.
why is machine learning attractive in real world problem? (3)
- ✓ * Write down the components of learning system.
How can we characterize "learning to drive a car." (3)
- ✓ * What is supervised learning? How to learn decision tree using entropy and information gain of attributes? (2+4)
- * Distinguished generalization loss and empirical loss. Show the model selection using error rates of training and validation data for different size decision tree (2+2)
- * What is univariate linear regression?
How to minimize the loss using gradient descent for fitting linear regression? (2+2)
- * What is the necessity of k-fold cross-validation

learning is essential for unknown environment

▣ learning: learning is an agents that can improve their behavior through diligent study of their own experience.

▣ Why would we want an agent to learn?

There are main three reason

1. First the designers can't anticipate all possible solutions that the agent might find itself.
2. Second the designers can't anticipate all changes over the time.
3. Third, sometimes human program have no idea how to program a solution themselves.

2 ▣ The capacity of a machine to imitate intelligent human behavior.

⇒ attractive / important

1. allows software application to become more accurate at predicting outcome without being explicitly programmed to do so.
2. Improve behavior through diligent study of their own experience.

Ex: Facebook, google, uber

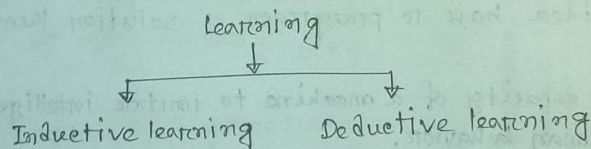
• make machine learning a central part of their operation

- 3 []
1. Performance Element
 2. Action in Environment
 3. Sensor to give I/p from Environment
 4. Environment
 5. Learning element
 6. Performance element.

Self driving car can be used

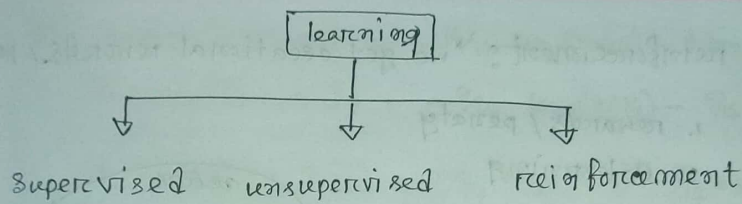
1. Bayesian regression algorithm
2. Neural Network " "
3. Decision Forest " "

4/4 []

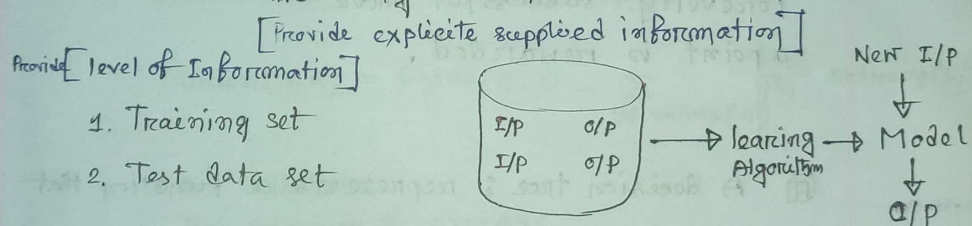


[] Inductive learning: A general function or rule from specific input output pairs.

[] Deductive learning: From a known general rule to a new rule that is logically entailed. . . it is useful because it allows more efficient processing.



☐ supervised: we get correct answers for each training instance.

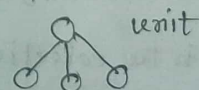
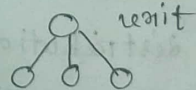


classification

regression

☐ unsupervised: we don't know anything

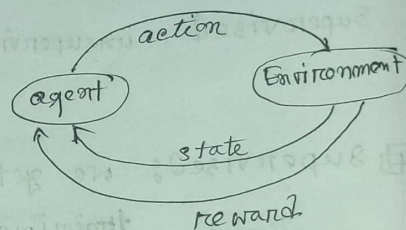
1. only input I/P
2. clustering $\rightarrow$ {group} $\leftarrow$ some category
3. k-mean algorithm.



The agent learns patterns in the input even though no explicit feedback is supplied.

reinforcement: "we get occasional rewards / punishment"

1. rewards / penalty
2. Q-learning



Ex: game playing

Forever successful move
a point is rewarded.

A decision tree: represents a function that takes as input a vector of attribute values and returns a "decision" - a single output value.

Entropy: Entropy measures homogeneity of examples [Entropy is measure of the uncertainty of a random variable]
Entropy measures homogeneity of examples that means entropy measures the impurity of a collection of examples. It depends from the distribution of the random variable.

$$\text{Entropy}(S) = -P_+ \log_2 P_+ - P_- \log_2 P_-$$

S is the collection of training examples

P_+ is the proportion of positive examples in S

P_- is the proportion of negative examples in S .

বিঃদ্রঃ - ধনাত্মক / ঋণাত্মক হলে Entropy = 0
 কারণ n/n হলে Entropy = 1.
 বাকি ক্ষেত্রে calculation করে হবে।

$$\text{Entropy}(14+, 0-) = -14/14 \log_2(14/14) - 0 \log_2(0) = 0$$

$$\text{Entropy}(9+, 5-) = -9/14 \log_2(9/14) - 5/14 \log_2(5/14) = 0.94$$

$$\text{Entropy}(7+, 7-) = -7/14 \log_2(7/14) - 7/14 \log_2(7/14) = 1.$$

Information gain: Information gain measures the expected reduction in entropy.

$$\text{Gain}(S, A) = \text{Entropy}(S) - \sum_{v \in \text{values}(A)} \frac{|S_v|}{|S|} \text{Entropy}(S_v)$$

A = attribute

S = collection of examples

Values (wind) = weak, strong

S = [9+, 5-]

S_{weak} = [6+, -2]

S_{strong} = [3+, 3-]

$$\text{gain}(S, \text{wind}) = \text{Entropy}(S) - \sum_{v \in \text{values}(A)} \frac{|S_v|}{|S|} \text{Entropy}(S_v)$$

$$= \text{Entropy}(S) - \sum_{v \in \{\text{weak}, \text{strong}\}} \frac{|S_v|}{|S|} \text{Entropy}(S_v)$$

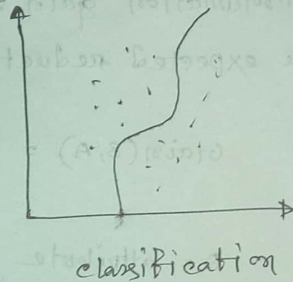
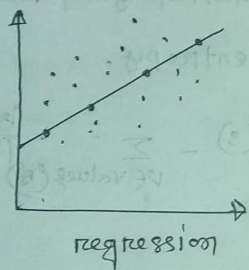
$$= \text{Entropy}(S) - (8/14) \text{Entropy}(S_{\text{weak}}) - (6/14) \text{Entropy}(S_{\text{strong}}).$$

Regression vs classification

Regression and classification are supervised learning

Both algorithm are used for prediction in

Machine learning and work with the labeled datasets



✓ regression: Regression algorithm used to predict the continuous value such as price, salary, age etc.

✓ classification: classification algorithms are used to predict discrete value such as Male, Female, True, False etc.

overfitting: over trained the training data.

7. **K-FOLD CROSS-VALIDATION**: [We can squeeze more out of data and still get an accurate estimate] using technique called k-fold cross validation.

The idea ~~of~~ is that each example serves double duty as training data and test data.

First we split the data into equal subset. Then we perform round of learning. on each round $\frac{1}{k}$ of the data is held out as a test set. and the remaining examples are used to training data. The average test set score of the rounds should then be a better estimate than a single score.

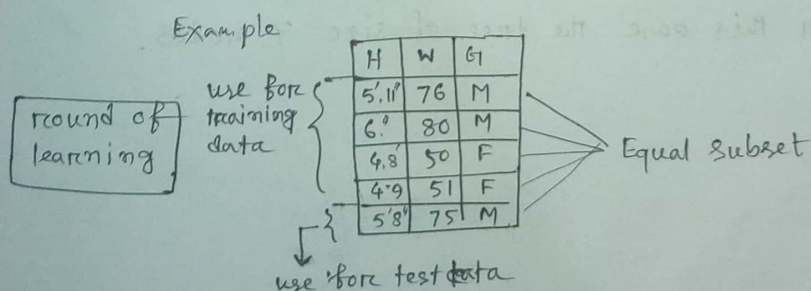
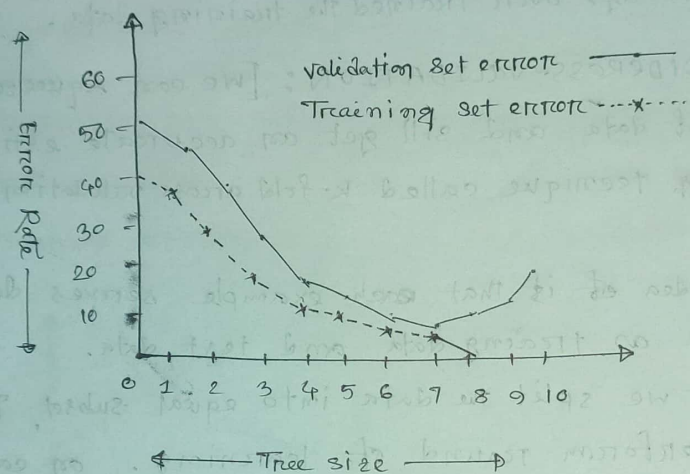


Fig: k-fold cross-validation.

5.



Error rates on training data and validation data (upper, solid line) for different size of decision trees. We stop when the training set error rate asymptotes, and then choose the tree with minimal error on the validation set. in this case the tree of size 7 nodes.

1	0	1
1	1	0
0	1	1
1	0	1
0	0	1
1	1	0
0	1	1
1	0	1

linear regression: The task of finding the h_w that best fit these data is called linear regression.

6. # univariate linear regression:

univariate linear regression refers to a linear regression model where we use only one independent variable x to learn a linear function that maps x to our dependent variable

univariate linear function

$$y = w_1 x + w_0$$

where w_1 and w_0 are real valued coefficients to be learned.

$$h_w(x) = w_1 x + w_0$$

$$\text{loss}(h_w) = \sum_{j=1}^N (y_j - (w_1 x_j + w_0))^2$$

w_1 and w_0 are obtained by minimizing the sum of the square of the errors.

$$\frac{\partial}{\partial \omega_0} \sum_{j=1}^N (y_j - (\omega_1 x_j + \omega_0))^2 = 0$$

$$\omega_0 = \frac{(\sum y_j - \omega_1 (\sum x_j))}{N}$$

$$\frac{\partial}{\partial \omega_1} \sum_{j=1}^N (y_j - (\omega_1 x_j + \omega_0))^2 = 0$$

$$\omega_1 = \frac{N (\sum x_j y_j) - (\sum x_j) (\sum y_j)}{N (\sum x_j^2) - (\sum x_j)^2}$$

We are trying to minimize the loss we will use gradient descent.

We choose any starting point in weight space - here a point (ω_0, ω_1) plane and then we move to a neighbouring point that is downhill.

repeat until we converge on the minimum possible loss:

$w \leftarrow$ any point in the parameter space.

loop until convergence do

for each w_i in w do

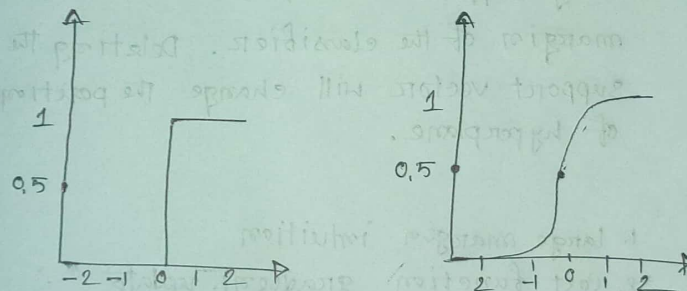
$$w_i \leftarrow w_i - \alpha \frac{\partial}{\partial w_i} \text{loss}(w)$$

the parameter α is called step size
and usually it called learning rate.

Multivariate linear regression: is nothing but more than multiple input.

- ✓ Hard Threshold
- ✓ Soft Threshold (logistic regression)

logistic function (z) = $\frac{1}{1 + e^{-z}}$ / sigmoid function



✓ Hard threshold

✓ Soft threshold

Hard thresholding is the process of setting to zero the coefficients whose absolute values are lower than the threshold.

Soft threshold is a processing tool that reduces the background in an image.

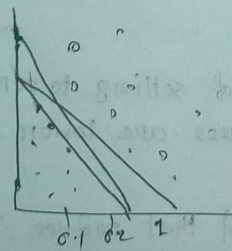
☐ separate data by using decision boundary

Support vector machine: is a type of deep learning algorithm that performs supervised learning for classification or regression of data group.

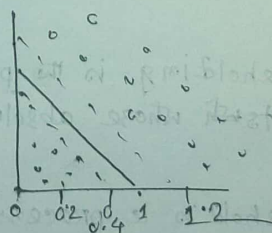
Effect of support vector machine:

using support vector, we maximize the margin of the classifier. Deleting the support vector will change the position of hyperplane.

1. large margin intuition
2. cost function gradient update
3. SVM implementation in python.

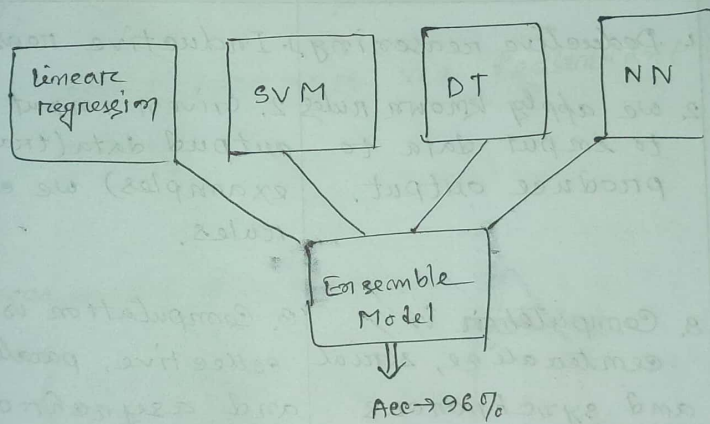


Three candidate linear separator.



Max margin separator

Ensemble learning: Ensemble learning is the process by which multiple models, such as classifiers or experts, are strategically generated and combined to solve a particular computational intelligence problem.



Neural Network

Md. Mahbubur Rahman Sir

Difference between Digital Computer and Neural Network

Digital Computer	Neural Network
1. Deductive reasoning.	1. Inductive reasoning.
2. we apply known rules to input data to produce output.	2. Given input and output data (training examples) we construct rules.
3. Computation is centralized, serial and synchronous.	3. Computation is collective, parallel and asynchronous.
4. Fast processing units	4. Slow processing units
5. Static infrastructure	5. Dynamic infrastructure
6. Reliable units	6. unreliable unit

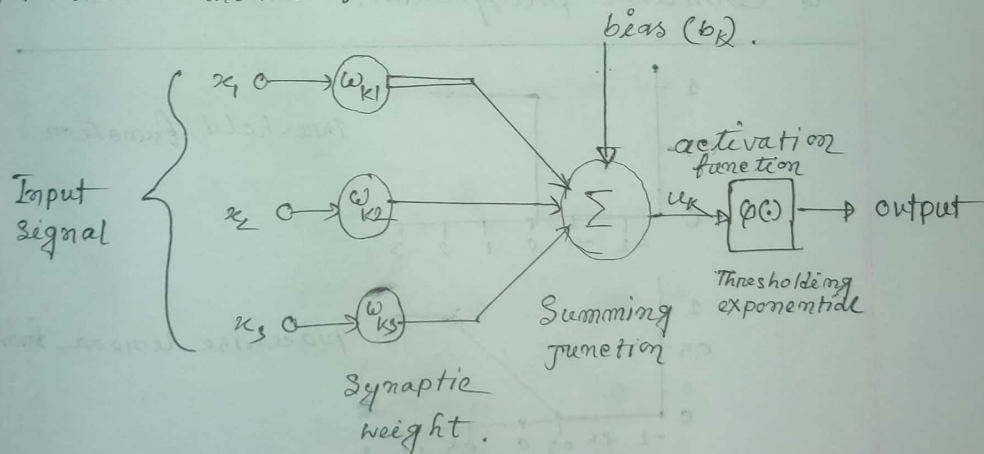
Why neural network

1. Adaptive learning
2. Self organization
3. Real time operation
4. Fault tolerance via Redundant Information coding.

Importance

1. non-linearity
2. I/o mapping
3. Adaptivity
4. contextual information
5. Fault tolerance
6. Implementability

Nonlinear model of a neuron



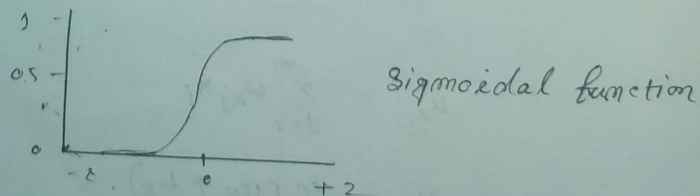
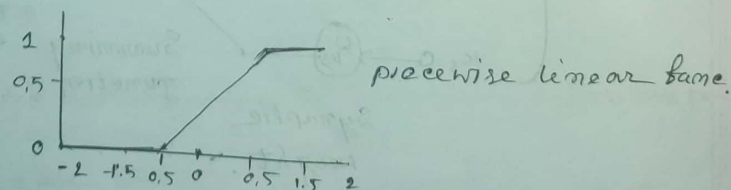
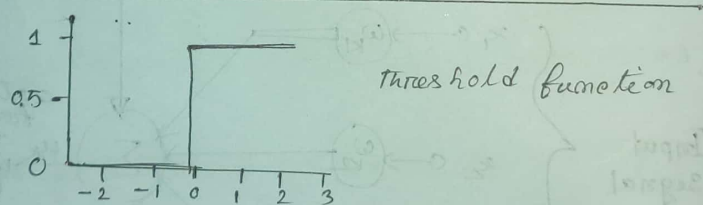
$$u_k = \sum_{j=1}^m w_{kj} x_j$$

$$y_k = \phi(u_k + b_k)$$

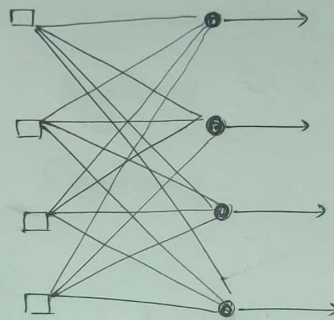
Artificial neural network: Artificial neural networks are computing systems inspired by the biological neural networks that constitute animal brains.

Artificial neural network also called neural network.

An artificial neural network is either a hardware implementation or a computer program.



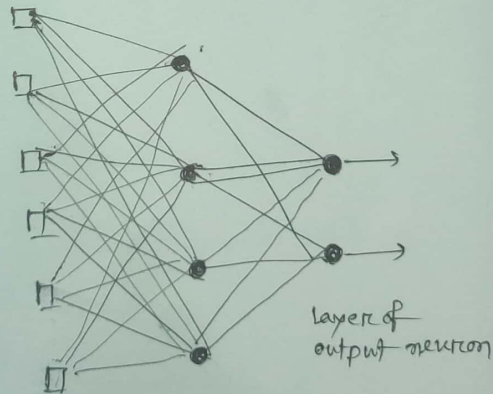
single layer Feedforward Network:



Input layer
of source node

output layer
of neurons

Multilayer feed forward network:



Input layer of
source node

layer of hidden
neurons

layer of
output neuron

Artificial Intelligence and neural networks: 2023

knowledge representation

learning

Reasoning

Simple model of machine learning:

